Deep Blue Sea

The ocean has layers, like floors in a very tall building. Each layer has different light, temperature, pressure, and food, and the animals that live there have special body parts and behaviors to survive.

My name: _____

My partner: _____

Our meetings

Lesson	Date	What we'll do
Lesson 1	Fri Oct 9	Dive! Build the Ocean Column
Lesson 2	Fri Nov 13	Sunlight and Twilight Zones
Lesson 3	Fri Jan 8	Into the Dark: Midnight, Abyss, and Trenches
Lesson 4	Fri Feb 5	Up at Night, Down by Day: Daily Migration
Lesson 5	Fri Mar 5	Practice Tournament
Lab I1	Before Lesson 3	Coast to Open Ocean: Tides, Waves, and Drying Out (home or an open Friday)
Lab I2	Before Lesson 5	Zone Detective: Invent a Deep-Sea Animal and Test Yourself (home or an open Friday)

At the tournament

A 50-minute event where partners answer questions and response sheets about ocean zones, adaptations, daily vertical migration, and representative animals from pictures and materials the leaders provide; teams bring only pencils, no notes.

Keep every handout for this event behind this divider, in order. Bring this binder to every Field B Fridays meeting.

Deep Blue Sea: What I can do

By tournament day, you should be able to do everything on this list. Check **Practicing** when you've started, and **Got it!** when you can do it without help. The lesson column tells you which handout to look back at: L = a group meeting, I = an independent lab you do at home or on an open Friday.

What I can do	Lesson	Practicing	Got it!
I can name the 5 depth zones in order with their nicknames and depths (0-200 m, 200-1,000 m, 1,000-4,000 m, 4,000-6,000 m, 6,000 to about 11,000 m).	L1 L2 L3 L5	☐	☐
I can compare the zones by sunlight, temperature, water pressure, and food.	L1 L3 L4 L5 I2	☐	☐
I can explain why sunlight fades with depth and why red light disappears first, so red animals are hidden in the deep.	L2	☐	☐
I can describe the sunlight zone and its animals (clownfish, sea turtles, tuna, dolphins, jellyfish, plankton) and explain photosynthesis and countershading.	L2 L5	☐	☐
I can describe the twilight zone and its animals (lanternfish, hatchetfish, squid, swordfish, octopus, crab) and explain big eyes and counter-illumination.	L2 L5	☐	☐
I can describe the midnight zone and its animals (anglerfish, gulper eel, viperfish, squid) and how they find food in the dark.	L3 L5	☐	☐
I can describe the abyss and the trenches and their animals (sea cucumbers, brittle stars, tripod fish, sea stars, shrimp, amphipods, snailfish, tube worms, crabs).	L3 L5	☐	☐
I can explain how deep-sea animals handle cold, high pressure, and little food (soft watery bodies, slow bodies, big mouths, stretchy stomachs, marine snow, bacteria partners).	L3 L4 I2	☐	☐
I can explain bioluminescence and 4 ways animals use it: to lure prey, to hide, to startle predators, and to find mates.	L2 L3 L4 I2	☐	☐
I can describe where the intertidal, neritic, and oceanic zones are and what makes each one different.	L5 I1	☐	☐
I can explain how intertidal animals (crabs, barnacles, sea stars, mussels) survive waves, tides, and drying out.	L5 I1	☐	☐
I can explain daily vertical migration: which animals move, which way, when, and why.	L4 L5	☐	☐
I can look at an animal I have never seen and predict its zone from its features.	L4 L5 I2	☐	☐
I can answer from memory with no notes, using flashcards and quizzing to practice.	L1 L5 I1 I2	☐	☐

Dive! Build the Ocean Column

TODAY I CAN

- I can name the 5 depth zones in order with their nicknames and depths (0-200 m, 200-1,000 m, 1,000-4,000 m, 4,000-6,000 m, 6,000 to about 11,000 m).
- I can compare the zones by sunlight, temperature, water pressure, and food.
- I can answer from memory with no notes, using flashcards and quizzing to practice.

The open ocean has 5 layers called depth zones. Pelagic (pe-LAJ-ik) means open ocean. You cannot bring notes to the tournament, so learn this chart by heart.

Today's steps: run it yourselves (22 minutes)

You need: 2 paper strips (115 cm), Model ocean strip, Meter stick, 12 animal photo cards per pair, Lesson 1 answer card. Set the timer for each step.

1. Both 2 min. Set the timer for 2 minutes. Read 'The 5 depth zones' chart out loud, taking turns. Say the memory trick: Every Mermaid Brings A Hat.
2. Both 8 min. Set the timer for 8 minutes. Write SURFACE at the top of your strip. Use the 'Paper model scale' box: always measure from the top and draw lines at 2, 10, 40, 60, and 110 cm. Label each zone and color it light blue to black.
3. Both 5 min. Set the timer for 5 minutes. Put each of the 12 animal cards on the zone where you think it lives. Use clues like color, eye size, and mouth size.
4. Event Captain 2 min. Read the animal guess answers from the answer card. Pairs move any wrong cards and say what clue fooled them.
5. Partner A 2 min. Set the timer for 2 minutes. Point to a zone on the strip; Partner B says its name and nickname. Switch halfway.
6. Both 3 min. Set the timer for 3 minutes. Answer the 3 'Check yourself' questions without looking at the chart. Then put the page in your divider.

Check your work: Ask your Event Captain for the answer card and check your 'Check yourself' answers. Fix any you missed in a different color, then compare your strip to the model strip.

The 5 depth zones (1 meter is about 3 feet)

Zone (nickname)	Depth	Light	Temperature	Pressure
Epipelagic (sunlight)	0 to 200 m	bright; plants and algae grow	warmest; changes with weather	low
Mesopelagic (twilight)	200 to 1,000 m	dim blue, fading	drops fast to about 39 F	about 20 to 100 times the surface
Bathypelagic (midnight)	1,000 to 4,000 m	no sunlight; only animal light	about 39 F	about 100 to 400 times
Abyssopelagic (abyss)	4,000 to 6,000 m	no sunlight	near freezing, about 36 F	about 400 to 600 times
Hadalpelagic (trenches)	6,000 m to about 11,000 m	no sunlight	near freezing	over 1,000 times

Paper model scale: 1 cm = 100 m (measure from the top)

2 cm	bottom of the sunlight zone (200 m)
10 cm	bottom of the twilight zone (1,000 m)
40 cm	bottom of the midnight zone (4,000 m)
60 cm	bottom of the abyss (6,000 m)
110 cm	deepest trench (about 11,000 m)

Memory trick for the order

E-M-B-A-H	Every Mermaid Brings A Hat: Epi, Meso, Bathy, Abysso, Hadal

Check yourself

1. Write the 5 depth zones from top to bottom.

2. Which zone gets enough sunlight for plants and algae to grow?

3. As you go deeper, what happens to light, temperature, and pressure?

At home before next time: Make 5 flashcards (15 minutes). Front: the zone's science name. Back: nickname, depth, and one animal. Punch a hole in the corner and put them on a ring. Quiz yourself until you get all 5 right twice, then have someone at home quiz you. Then label the ocean column in your Ocean Explorer Field Journal (p. 3) for the Zone Diver badge. Bring your cards and journal to Lesson 2 on Nov 13. Museum trip (Sat Nov 7, meet 10 am at the Daily Planet Cafe): bring your journal, find 2 ocean animals, and fill in the Museum Day page (p. 17) with the zone you think each lives in and the clue that told you.

Sunlight and Twilight Zones

TODAY I CAN

- I can name the 5 depth zones in order with their nicknames and depths (0-200 m, 200-1,000 m, 1,000-4,000 m, 4,000-6,000 m, 6,000 to about 11,000 m).
- I can explain why sunlight fades with depth and why red light disappears first, so red animals are hidden in the deep.
- I can describe the sunlight zone and its animals (clownfish, sea turtles, tuna, dolphins, jellyfish, plankton) and explain photosynthesis and countershading.
- I can describe the twilight zone and its animals (lanternfish, hatchetfish, squid, swordfish, octopus, crab) and explain big eyes and counter-illumination.
- I can explain bioluminescence and 4 ways animals use it: to lure prey, to hide, to startle predators, and to find mates.

Sunlight fades as you go down. Animals in the top 2 zones use light, color, and big eyes to eat and hide. Study this page with flashcards; no notes at the tournament.

Today's steps: run it yourselves (22 minutes)

You need: Blue-covered flashlight, Red, blue, and black paper squares, 4 fish outlines (1 per student), 12 animal cards per pair, Lesson 2 answer card. Set the timer for each step.

1. Both 2 min. Set the timer for 2 minutes. Read the 'Sunlight zone' box. Talk: why do most ocean animals live near the top?
2. Both 5 min. Set the timer for 5 minutes. In a dim corner or under a coat, shine the blue flashlight on the red, blue, and black squares. Which one looks black? Water soaks up red light first. Answer 'Check yourself' questions 1 and 2.
3. Both 4 min. Set the timer for 4 minutes. Color a paper fish dark blue on top and white on the belly. Hold it over the dark floor, then up against a bright window. This is countershading. Answer question 3.
4. Both 7 min. Set the timer for 7 minutes. Read the 'Twilight zone' box. Sort the 12 cards into sunlight and twilight. Pick 4 animals and write one adaptation for each in the 'Adaptation match' table.
5. Both 2 min. Set the timer for 2 minutes. Answer question 4 from memory.
6. Event Captain 2 min. Read the answers from the answer card. Pairs fix mistakes and file the page. Take an Independent Lab I1 page.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Sunlight zone (epipelagic, 0-200 m)

Photosynthesis	plants and algae use sunlight to make food; only happens in this zone
Plankton	tiny drifters; phytoplankton (plant-like) and zooplankton (animals like jellyfish and baby fish)
Countershading	dark back, light belly (tuna, dolphins, sea turtles)
Animals	clownfish, sea turtles, tuna, dolphins, jellyfish, zooplankton

Twilight zone (mesopelagic, 200-1,000 m)

Light	dim blue, too dark for plants; red light is gone
Big eyes	catch every bit of dim light (squid, hatchetfish)
Bioluminescence	making light with chemicals in the body (bye-oh-loo-mih-NES-ens)
Counter-illumination	lights on the belly hide the fish's shadow from below (lanternfish, hatchetfish)
Red color	looks black, so it hides (deep-sea red crab)
Animals	lanternfish, hatchetfish, squid, swordfish, octopus, crab

Adaptation match

Animal	Zone	Adaptation and how it helps

Check yourself

1. Which color of light disappears first in the ocean?

2. Which color of light goes the deepest?

3. How does countershading help a tuna hide?

4. A hatchetfish has lights on its belly. What is this called and why does it help?

At home before next time: Do Independent Lab I1 (Coast to Open Ocean) before Lesson 3 on Jan 8. Also make 12 animal flashcards (15 minutes): front = animal name (draw it if you like); back = zone and one adaptation. Add them to your ring and quiz yourself. Field Journal: do the Adaptation Detective match (p. 6) and log 2 Ocean Animal pages (p. 7-12) from an aquarium, a book, or a video for the Adaptation Detective badge.

Into the Dark: Midnight, Abyss, and Trenches

TODAY I CAN

- I can name the 5 depth zones in order with their nicknames and depths (0-200 m, 200-1,000 m, 1,000-4,000 m, 4,000-6,000 m, 6,000 to about 11,000 m).
- I can compare the zones by sunlight, temperature, water pressure, and food.
- I can describe the midnight zone and its animals (anglerfish, gulper eel, viperfish, squid) and how they find food in the dark.
- I can describe the abyss and the trenches and their animals (sea cucumbers, brittle stars, tripod fish, sea stars, shrimp, amphipods, snailfish, tube worms, crabs).
- I can explain how deep-sea animals handle cold, high pressure, and little food (soft watery bodies, slow bodies, big mouths, stretchy stomachs, marine snow, bacteria partners).
- I can explain bioluminescence and 4 ways animals use it: to lure prey, to hide, to startle predators, and to find mates.

Below 1,000 meters there is no sunlight at all. It is cold, the water pressure is huge, and food is hard to find. Learn these with flashcards.

Today's steps: run it yourselves (27 minutes)

You need: 2 syringes with marshmallows (no needles), Marine snow confetti, 3 pom-poms, and a tray, 2 spoons and 2 large cups, 10 deep animal cards per pair, Lesson 3 answer card. Set the timer for each step.

1. Both 4 min. Set the timer for 4 minutes. Read the 'Deep zones' box and 'Marine snow' out loud. Talk: without sunlight, where does food come from?
2. Both 5 min. Set the timer for 5 minutes. Cover the syringe tip with a finger and push the plunger. Watch the marshmallow shrink, then pull back. Read 'Soft, watery body' in 'How they survive'.
3. Event Judge 6 min. Spread the confetti and pom-poms on the tray. One partner uses a spoon (small mouth), the other a big cup (gulper eel). Eyes closed, 10 seconds each, one scoop per second. Count what each caught. Why do deep fish have huge mouths?
4. Both 7 min. Set the timer for 7 minutes. Match each animal card to its zone and adaptation using 'How they survive'. Fill in the 'Deep animal match' table.
5. Both 3 min. Set the timer for 3 minutes. Answer the 3 'Check yourself' questions from memory.
6. Event Captain 2 min. Read the answers from the answer card. Pairs fix mistakes and file the page.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Deep zones

Midnight (bathypelagic, 1,000-4,000 m)	total dark except animal light, about 39 F. Anglerfish, gulper eel, viperfish, squid.
Abyss (abyssopelagic, 4,000-6,000 m)	near freezing, huge pressure, flat muddy sea floor. Sea cucumbers, brittle stars, tripod fish, sea stars, shrimp.
Trenches (hadalpelagic, 6,000 to about 11,000 m)	deep cracks in the sea floor; pressure over 1,000 times the surface. Amphipods, snailfish, tube worms, sea stars, crabs.

How they survive

Marine snow	bits of dead plankton, poop, and dead animals drifting down; main food in the deep
Big mouth, stretchy stomach	swallow any food that comes by, even big prey (gulper eel, anglerfish)
Lure	a glowing bait on the head (anglerfish)
Soft, watery body	no air pockets, so pressure does not crush it (snailfish)
Slow body	uses little energy, so it needs little food
Dark or red color	invisible in the dark

Bacteria partners	tube worms get food from bacteria that use chemicals, not sunlight (chemosynthesis)
4 uses of animal light	lure prey, hide, startle predators, find mates

Deep animal match

Animal	Zone	Adaptation
anglerfish		
gulper eel		
viperfish		
tripod fish		
sea cucumber		
snailfish		
tube worm		

Check yourself

1. What is marine snow?

2. Why does deep water pressure not crush a snailfish?

3. Name 4 ways animals use bioluminescence.

At home before next time: Make 10 more flashcards for the deep animals (front: animal; back: zone and adaptation). Then play 'Zone Shuffle' with someone at home: they read the back of a card and you guess the animal. Do all your cards (zones plus animals) twice this week. Field Journal: Glow Hunt (p. 13), finding an animal for each of the 4 ways to use light, for the Glow Hunter badge.

Up at Night, Down by Day: Daily Migration

TODAY I CAN

- I can compare the zones by sunlight, temperature, water pressure, and food.
- I can explain how deep-sea animals handle cold, high pressure, and little food (soft watery bodies, slow bodies, big mouths, stretchy stomachs, marine snow, bacteria partners).
- I can explain bioluminescence and 4 ways animals use it: to lure prey, to hide, to startle predators, and to find mates.
- I can explain daily vertical migration: which animals move, which way, when, and why.
- I can look at an animal I have never seen and predict its zone from its features.

Diel (DIE-el) means daily. Diel vertical migration is when ocean animals swim up and down every day. It is the biggest animal movement on Earth.

Today's steps: run it yourselves (27 minutes)

You need: DAY card and NIGHT card, 20 green food squares, 4 mystery animal cards, Speed quiz cards (8), Lesson 4 answer card. Set the timer for each step.

1. Team 2 min. Set the timer for 2 minutes. Read 'Daily vertical migration' out loud. Pick jobs: 2 lanternfish, 1 tuna, 1 scientist who flips the card and keeps score.
2. Team 9 min. Play 4 rounds. Lanternfish start below the 200 m line. Flip DAY or NIGHT. DAY: the tuna tags any lanternfish above 200 m. NIGHT: the tuna closes its eyes and lanternfish go up for food. Switch jobs each round.
3. Both 4 min. Set the timer for 4 minutes. Talk: when is it safe to go up, and why go up at all? Answer 'Check yourself' question 1.
4. Event Judge 7 min. Read each mystery animal card. Pairs write the zone and clue in 'Predict the zone', then flip the card to check. Then pairs do 'Match the adaptation to its job' and question 2.
5. Event Judge 3 min. Read the speed quiz cards, one at a time. Pairs write each answer on the whiteboard and hold it up. No notes.
6. Event Captain 2 min. Read the answers from the answer card. Pairs fix mistakes and file the page.

Check your work: Ask your Event Captain for the answer card and check your answers. Fix any you missed in a different color.

Daily vertical migration

Who	zooplankton like krill and copepods, lanternfish, squid, and the animals that eat them
Night	swim up toward the surface to eat plankton
Day	swim back down to the dark twilight zone to hide
Why	food is near the surface, but daylight lets predators see them; the dark keeps them safe
Bonus	the moving animals carry food and carbon from the surface down into the deep

Predict the zone

Mystery animal	Zone	Clue that told me
A		
B		
C		
D		

Match the adaptation to its job

big eyes	hold on in waves (mussel)

belly lights	catch dim light (twilight)
huge mouth	handle high pressure (trenches)
soft watery body	eat rare food (midnight)
countershading	dark top, light bottom to hide (sunlight)
strong threads	hide shadow from below (twilight)

Check yourself

1. Why do lanternfish go up at night and not in the day?

2. How do a slow body and a big mouth help an animal in the dark?

At home before next time: Do Independent Lab I2 (Zone Detective) before Lesson 5 on Mar 5. It has a practice test, so do it without notes. Use the Creature Creator page in your Field Journal (p. 15) to design your animal (Creature Creator badge). Also make 5 flashcards for migration words (diel vertical migration, zooplankton, krill, marine snow, counter-illumination) and add them to your ring, then draw day and night migration on page 14 (Migration Tracker badge). Next time is a practice tournament.

Practice Tournament

TODAY I CAN

- I can name the 5 depth zones in order with their nicknames and depths (0-200 m, 200-1,000 m, 1,000-4,000 m, 4,000-6,000 m, 6,000 to about 11,000 m).
- I can compare the zones by sunlight, temperature, water pressure, and food.
- I can describe the sunlight zone and its animals (clownfish, sea turtles, tuna, dolphins, jellyfish, plankton) and explain photosynthesis and countershading.
- I can describe the twilight zone and its animals (lanternfish, hatchetfish, squid, swordfish, octopus, crab) and explain big eyes and counter-illumination.
- I can describe the midnight zone and its animals (anglerfish, gulper eel, viperfish, squid) and how they find food in the dark.
- I can describe the abyss and the trenches and their animals (sea cucumbers, brittle stars, tripod fish, sea stars, shrimp, amphipods, snailfish, tube worms, crabs).
- I can describe where the intertidal, neritic, and oceanic zones are and what makes each one different.
- I can explain how intertidal animals (crabs, barnacles, sea stars, mussels) survive waves, tides, and drying out.
- I can explain daily vertical migration: which animals move, which way, when, and why.
- I can look at an animal I have never seen and predict its zone from its features.
- I can answer from memory with no notes, using flashcards and quizzing to practice.

Practice tournament: no notes! Read each station card and write your answers here. When time is called, move on.

Today's steps: run it yourselves (27 minutes)

You need: 8 station cards (1-8), Lesson 5 answer card. Set the timer for each step.

1. Event Judge 1 min. Collect all flashcards and notes. Set out the 8 station cards in order. Read aloud: 'No notes. 3 minutes per station. Move to the next number when time is called.' Pairs start at Stations 1 and 5.
2. Event Judge 12 min. Set the timer for 3 minutes each round for 4 rounds. Call 'One minute left' and 'Rotate!'. Write every answer on the 'Answer sheet'.
3. Event Judge 12 min. Keep going for 4 more rounds so every pair visits all 8 stations. Write something for every question: a guess can earn points.
4. Event Captain 2 min. Read the answers from the answer card. Pairs score their sheet and fill in 'My review plan'.

Check your work: Ask your Event Captain for the answer card and score your answer sheet. Circle 2 topics to study and write them in 'My review plan'.

Answer sheet

Station	Answer
1	
2	
3	
4	
5	
6	
7	
8	

Before the tournament

- [ ] I can name all 8 zones and say the depth of the 5 depth zones.
- [ ] I can name 2 animals from each zone and one adaptation for each.
- [ ] I can explain daily vertical migration and bioluminescence.
- [ ] I bring only pencils. No notes allowed.

My review plan

1. Which 2 topics will I study before the tournament?

At home before next time: Study your flashcards for 10 minutes every other day until the tournament. Put the cards you miss in a 'practice' pile and do that pile first each time. Ask someone to give you the practice stations again from memory. Field Journal: cover the zone tags on the Animal Collector pages (p. 4-5) and name 10 animals' zones in under 2 minutes for the Speed Spotter badge.

Lab I1: Coast to Open Ocean: Tides, Waves, and Drying Out

THIS LAB HELPS ME

- I can describe where the intertidal, neritic, and oceanic zones are and what makes each one different.
- I can explain how intertidal animals (crabs, barnacles, sea stars, mussels) survive waves, tides, and drying out.
- I can answer from memory with no notes, using flashcards and quizzing to practice.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. Scientists also split the ocean by how far it is from land. Picture walking from the beach out to the open sea. In this lab you test how shore animals hang on in waves and stay wet at low tide.

Steps (28 minutes)

You need: 2 paper towel squares, a lidded container, and a plate, Bowl or tub, 2 bottle caps, string, a spoon or rock, a suction cup (if you have one), Towel, 7 index cards. Set the timer for each step.

1. Me 2 min. Set the timer for 2 minutes. Wet 2 paper towel squares the same amount. Put one in the closed container (a barnacle with its plates shut) and one on the open plate (plates open). Put both in a warm spot and write the time on the 'Drying test' line.
2. Me 8 min. Set the timer for 8 minutes. Read 'Zones from the shore out'. In the 'Side view' box, draw the beach, the high and low tide lines, the flat continental shelf, and the drop off. Label intertidal, neritic, and oceanic. Where do the 5 depth zones go?
3. Me 6 min. Set the timer for 6 minutes. Put 3 cm of water in the bowl. Tie one cap to the spoon or rock with the string (a mussel with threads). Set the other cap loose next to it, and press the suction cup to the bottom if you have one. Make waves with your hand for 30 seconds. Fill in the 'Wave test' table.
4. Me 5 min. Set the timer for 5 minutes. Read 'Intertidal survival'. Then do 'Which zone am I?': circle the zone for each clue.
5. Me 4 min. After at least 1 hour (or at the end of your open Friday), feel both towels. Which is wetter? Finish the 'Drying test' line. Dump the water and wipe up.
6. Me 3 min. Set the timer for 3 minutes. Answer the 'Check yourself' questions without looking at the boxes. Then make 7 flashcards: intertidal, neritic, oceanic, tide, barnacle, mussel, sea star.

Check your work: Check 'Which zone am I?' and 'Check yourself' against the 'Zones from the shore out' and 'Intertidal survival' boxes. On an open Friday or at Lesson 3, ask your Event Captain for the Lab I1 answer card and fix any you missed in a different color.

Zones from the shore out

Intertidal	the shore between high tide and low tide. Covered by water, then left dry, twice a day. Waves, sun, and changing temperature and saltiness. Animals: crabs, barnacles, sea stars, mussels.
Neritic	shallow water over the continental shelf (the flat edge of a continent), to about 200 m deep. Lots of sunlight and nutrients, so it is full of life: kelp, coral reefs, most fishing.
Oceanic	the open ocean past the continental shelf. Deep water; all 5 depth zones are under it.
Tide	the daily rise and fall of the sea

Intertidal survival

Barnacles	glued head-down to rocks; close their plates at low tide to stay wet
Mussels	hold on with tough threads (byssal threads); shut tight at low tide
Sea stars	hundreds of tube feet work like suction cups to grip rocks and pry open mussels
Crabs	hard shell; hide under rocks and seaweed; scuttle to follow the water

Side view: draw and label

Beach, high tide line, low tide line, continental shelf, drop off, intertidal, neritic, oceanic

Wave test

What I used	Real animal it acts like	Did it hold on?
Cap tied with string	Mussel (byssal threads)	
Suction cup	Sea star (tube feet)	
Loose cap	Shell with nothing holding it	

Drying test

Start time: _____ Check time: _____ The wetter towel was the one on the _____ (plate / in the container). This is like a barnacle that _____ at low tide.

Which zone am I?

1. Twice a day I am covered by water, then left dry.
◯ Intertidal ◯ Neritic ◯ Oceanic

2. I am shallow water over the continental shelf, full of fish and kelp.
◯ Intertidal ◯ Neritic ◯ Oceanic

3. I am the open ocean past the shelf, with all 5 depth zones below me.
◯ Intertidal ◯ Neritic ◯ Oceanic

4. Barnacles and mussels live here.
◯ Intertidal ◯ Neritic ◯ Oceanic

5. Most of the world's fishing happens here.
◯ Intertidal ◯ Neritic ◯ Oceanic

6. Tuna and whales cross me far from land.
◯ Intertidal ◯ Neritic ◯ Oceanic

7. Waves pound me and the sun dries me out.
◯ Intertidal ◯ Neritic ◯ Oceanic

8. My water is less than about 200 m deep and gets lots of sunlight and nutrients from land.
◯ Intertidal ◯ Neritic ◯ Oceanic

Check yourself

1. How do 2 intertidal animals survive low tide?

2. Which zone has the most fish for people to catch? Why?

3. How do mussels and sea stars hold on when waves crash?

Bring to the next meeting: Bring this page and your 7 new flashcards (on your ring) to Lesson 3 on Jan 8. Teach someone at home the difference between the 3 shore-to-sea zones and the 5 depth zones using your paper ocean column.

Lab I2: Zone Detective: Invent a Deep-Sea Animal and Test Yourself

THIS LAB HELPS ME

- I can compare the zones by sunlight, temperature, water pressure, and food.
- I can explain how deep-sea animals handle cold, high pressure, and little food (soft watery bodies, slow bodies, big mouths, stretchy stomachs, marine snow, bacteria partners).
- I can explain bioluminescence and 4 ways animals use it: to lure prey, to hide, to startle predators, and to find mates.
- I can look at an animal I have never seen and predict its zone from its features.
- I can answer from memory with no notes, using flashcards and quizzing to practice.

Where: Home or an open Friday. You can do this alone or with your partner, with no adult needed. At the tournament you will see animals you have never met. The trick is to read the body clues. In this lab you predict zones, invent your own deep-sea animal, and take a no-notes practice test.

Steps (28 minutes)

You need: This page, Colored pencils, Your flashcard ring (keep it closed until step 5). Set the timer for each step.

1. Me 3 min. Set the timer for 3 minutes. Read the 'Zone clues' box out loud.
2. Me 8 min. Set the timer for 8 minutes. Read each animal in 'Mystery animals'. Write its zone and the clue that told you. Do not look at your flashcards.
3. Me 8 min. Set the timer for 8 minutes. Pick the midnight zone, the abyss, or a trench. Draw your own animal in the 'Invent a deep-sea animal' box. Label 3 adaptations: one for the dark, one for cold or pressure, and one for finding food. Give it a name.
4. Me 6 min. Set the timer for 6 minutes. Fold the page so only the 'Practice test' shows. Answer all 10 from memory. If someone is home, they can read each question to you.
5. Me 3 min. Set the timer for 3 minutes. Check your test with your flashcards and your lesson pages. Write your score and list the cards you missed in 'My practice pile'.

Check your work: Check the practice test with your flashcards and lesson pages. On an open Friday or at Lesson 5, ask your Event Captain for the Lab I2 answer card to check 'Mystery animals'. Fix any you missed in a different color.

Zone clues

Sunlight	fast body, countershading (dark back, light belly), bright colors on reefs
Twilight	big eyes, belly lights, silver or red body
Midnight	black body, huge mouth, long teeth, glowing lure, stretchy stomach
Abyss	lives on the cold, muddy sea floor (4,000 to 6,000 m); slow; eats marine snow
Trenches	soft jelly body, no air pockets, deeper than 6,000 m
Intertidal	holds on to rocks; closes up or hides when the tide goes out

Mystery animals

#	Clues	Zone	Clue that told me
1	Blue-gray back, white belly, torpedo body; hunts near the surface		
2	Bright red body; lives at about 700 m where light is dim blue		
3	Long thin body, giant hinged jaw, stomach that stretches; it is totally dark		
4	Stands on 3 stiff fins on the muddy floor at 5,000 m, facing the current		
5	Pale pink, soft jelly-like body, no scales; lives at 8,000 m		
6	Tough threads anchor it to a rock; its shell snaps shut when the tide goes out		

Invent a deep-sea animal (name, zone, 3 labeled adaptations)

Practice test (no notes!)

1. Name the 5 depth zones in order.

2. Which zone has photosynthesis?

3. Which color of light disappears first?

4. What is marine snow?

5. Name 4 ways animals use bioluminescence.

6. When do lanternfish swim up toward the surface, and why?

7. Which zone is between high tide and low tide?

8. Which zone is shallow water over the continental shelf?

9. Why do deep fish have big mouths and stretchy stomachs?

10. Where do tube worms get their food?

My practice pile

1. My score: ____ out of 10. The flashcards I will practice most:

Bring to the next meeting: Bring your invented animal to show at the Lesson 5 Huddle. Study your practice pile first each day until then.